haptic navigation headband: brief

what we are building

a headband that helps a blind or low-vision person walk to a destination using vibration alone. no audio, so they can keep listening to traffic and people around them.

it does two things:

1. **tells you which way to turn.** if you're facing the right direction, nothing vibrates. if you need to go left, the left motor buzzes until you're facing the right way again, then it stops.
2. **warns you about things in front of you, and which way to step around them.** two time-of-flight sensors watch the space ahead. the front motor pulses faster the closer something gets, and if you need to step around it, the left or right motor gives a double-pulse toward whichever side is clearer.

it's meant to work alongside a cane, not replace one. a sensor at forehead height won't catch curbs, stairs or drops, since those need ground contact to detect, which nothing on this headband does. this stays a supplement to a cane or dog, not a replacement, until (if ever) we add and validate real drop-off detection with an orientation and mobility professional.

hardware

- **band:** inner ring (touches skin, holds the motors), a middle layer for wires and electronics, and an outer ring with cutouts for the sensors. traced as a 190 mm (front-to-back) by 145 mm (side-to-side) ellipse, about 1.5 to 2 in tall, with a ratchet dial to adjust for the 20-21 in+ circumference range.
- **three coin vibration motors:** front (forehead), left and right (temples). each driven through a transistor with a flyback diode.
- **two VL53L5CX time-of-flight sensors:** forward-facing, toed outward about 22° each (44° between them) so their zones meet near dead center with no gap. each is an 8×8 zone sensor, not a single point, which is what lets us tell which side of an obstacle is clearer.
- **ESP32:** runs everything and talks to the phone over bluetooth.
- **IMU with magnetometer (BNO055 or BNO085):** gives the current head heading. mounted at the back of the band, offset a few cm to one side of the battery, not directly on top of it, with its own twisted-pair wire run.
- **LiPo battery, USB-C charging board, power switch:** battery sits at the back of the band, opposite the sensors and motors, so the weight balances front to back instead of pulling forward.

software

getting the route onto the headband

maps API → JSON route → phone app → parse into an array of waypoints → write array to headband memory

while walking

1. headband takes the next waypoint in the array.
2. it compares the current position with that waypoint to get a target bearing.
3. it reads the magnetometer for the current heading.
4. delta = target bearing − heading, wrapped to ±180°.
5. if delta is within ±20°, no vibration. if it's past that, the left or right motor runs, stronger for bigger errors.
6. once you're within roughly 10 m of the waypoint, it moves to the next one. after the last waypoint, all three motors pulse twice to say you've arrived.

positioning: outdoors vs GPS-denied

phone streams GPS position to the headband over bluetooth roughly once a second. this is simpler than putting a GPS module on the band, and it means the phone can handle rerouting if someone takes a wrong turn.

GPS alone breaks down indoors, underground, and between tall buildings, exactly where a blind user most needs reliable guidance. for those cases we bring in the Hyperlocation SDK, which fuses IMU, magnetometer, barometer, and opportunistic signals (wifi, BLE, UWB) on-device to keep positioning working without GPS. practically: the phone (or headband, if we later add BLE/UWB hardware) runs Hyperlocation as a fallback source of position whenever GPS confidence drops, and the

navigation logic in step 2-5 above doesn't need to know which source the bearing came from, it just gets a target bearing either way.

obstacle detection and steering

- both sensors' 8×8 zone grids are read together as one wide grid.
- if the closest zone anywhere in that grid is under the warning distance (start at 100 cm, tune from testing), the front motor starts pulsing, faster as the closest reading gets closer.
- at the same time, split the combined grid into a left half and a right half, and take the minimum distance in each half. whichever half is more open is the side with more room to step toward.
- if one side is clearly more open than the other, that side's motor double-pulses (different rhythm from navigation's steady buzz) to say "step this way."
- if both sides are about equally blocked, all three motors pulse together fast, meaning stop, the path is blocked straight across.

keeping the vibrations tellable apart

this was the hard part, since we're reusing the left and right motors for two different jobs. the fix is rhythm, not new motors:

Meaning	motor(s)	rhythm
off course, turn this way	left or right	steady buzz
obstacle ahead	front	pulses, speed scales with distance
step this way around it	left or right	double-pulse, repeating
path fully blocked	all three	fast pulse together
nothing needed	none	silence

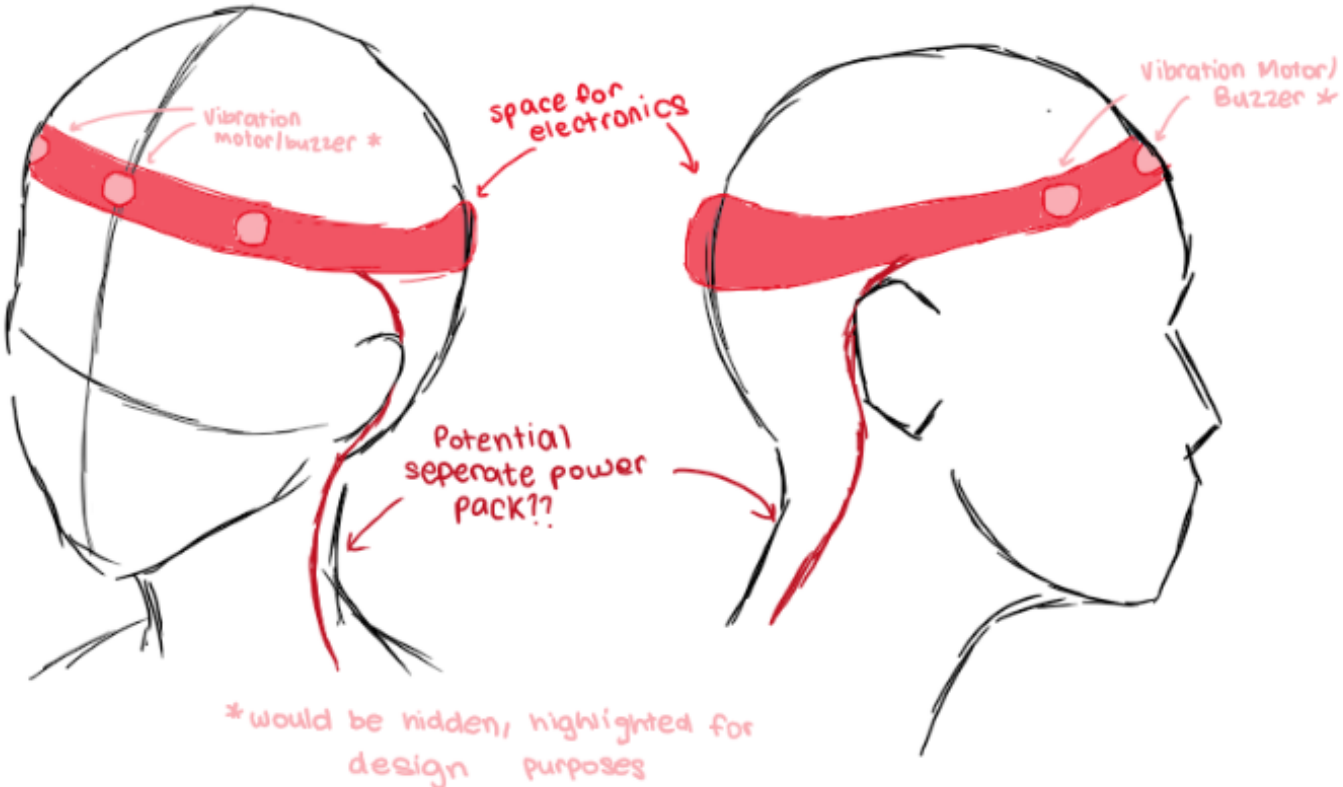
priority order: obstacle and step cues always override navigation. the moment an obstacle trips, the steady nav buzz pauses and the double-pulse steering cue takes over. once the obstacle clears, navigation picks back up from where it left off. nobody has to learn a new mapping, left still means left, right still means right, only the texture changes with what it means, the same way a phone's ringtone and notification buzz share a speaker but never get confused for each other.

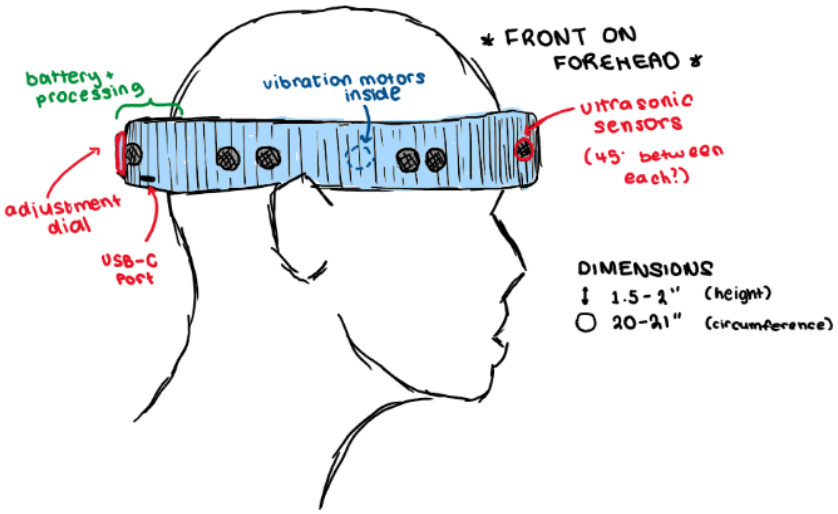
head heading vs body direction

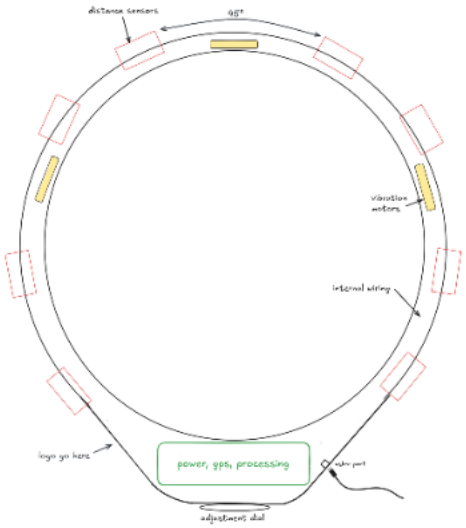
for v1, we're using head heading as a direct stand-in for walking direction; the deadband logic assumes the user is generally facing the way they're walking, since that's also how they'll interpret "turn until it stops." this breaks down if someone turns their head to listen to traffic or check a sound without turning their body, since the nav cue would shift with them. we're accepting this as a known v1 limitation rather than solving it with a second sensor, since the added chest or body-worn sensor complexity isn't worth it before we've validated the core system works. if testing shows this causes real confusion, the next version option is a second, body-worn heading reference (e.g. a chest clip) that the headband's cue is calculated against instead of the head's own compass.

known limitations

- motors and battery interfere with the magnetometer, both from their static magnets and from current draw while active. placement (IMU offset from battery, away from motor wiring) and a motors-on/motors-off calibration check handle most of this, but it needs bench testing before the position is locked in.
- GPS accuracy degrades to a few meters and worse between tall buildings; Hyperlocation covers indoor and GPS-denied cases but adds a second positioning system to test and validate.
- soft surfaces (clothing, bushes) absorb the sensor signal and angled flat surfaces can deflect it, so some obstacles will still be missed regardless of sensor choice.
- this is a navigation and obstacle-avoidance aid, not a full cane replacement, since it can't detect curbs, steps, or drop-offs at ground level.


Vibration motor/buzzer *
space for electronics
Vibration Motor/ Buzzer *
Potential seperate power pack??
*would be hidden, highlighted for design purposes


* FRONT ON FOREHEAD *
battery + processing
vibration motors inside
ultrasonic sensors
(45° between each?)
adjustment dial
USB-C Port
DIMENSIONS
1.5-2" (height)
20-21" (circumference)


distance sensors
45°
vibration motors
internal wiring
power, gps, processing
usb-c port
adjustment dial